



Alberta Public Health Disease Management Guidelines

Eastern Equine Encephalitis



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Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of Eastern Equine Encephalitis (EEE) virus from an appropriate clinical specimen (e.g., cerebrospinal fluid (CSF), serum)^(B)

OR

- Demonstration of EEE virus antigen in tissue, blood or CSF

OR

- Detection of EEE viral nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., plasma, CSF, serum)^(B)

OR

- Seroconversion or significant difference between acute and convalescent phase EEE Hemagglutination Inhibition Assay (HAI) titers ideally taken at least two weeks apart from an appropriate clinical specimen (e.g., serum)^(B) and confirmed by Plaque Reduction Neutralization Assay (PRNT)^(C)

Probable Case

Clinical illness^(A) and one of the following:

- Seroconversion or significant difference between acute and convalescent phase EEE HAI titres ideally taken at least two weeks apart but not confirmed by PRNT^(C)

OR

- Stable elevated serial HAI titres^(D) to EEE that occur during a period when and where arboviral transmission is likely^(E)

OR

- Single elevated HAI titre^(D) to EEE that occurs during a period when and where arboviral transmission is likely^(E)

^(A) Clinical illness is characterized by a febrile illness of variable severity associated with neurological symptoms ranging from headache to aseptic meningitis or encephalitis. Arboviral encephalitis cannot be distinguished clinically from other central nervous system (CNS) infections. Symptoms can include headache, confusion or other alteration in sensorium, nausea and vomiting. Signs may include fever, meningismus, cranial nerve palsies, paresis or paralysis, sensory deficits, altered reflexes, convulsions, abnormal movements and coma of varying degree.

^(B) Refer to the [National Microbiology Laboratory \(NML\) Guide to Services](#) for current specimen collection and submission information.

^(C) Seroconversion indicates recent infection with a flavivirus (e.g., dengue fever, California serogroup, West Nile virus or yellow fever) but cannot pinpoint to which one due to antibody cross-reactivity. These are considered probable cases. PRNT confirmatory testing may be requested at NML through the Public Health Laboratory (ProvLab).

^(D) A single elevated antibody titre of $\geq 1:40$ by HAI suggests recent infection. It is recommended that a second specimen be collected. A second laboratory result with a stable (unchanged/static or $\leq$ twofold rise) elevated antibody titre is suggestive of recent infection.

^(E) EEE exposure is most likely to occur during the summer and early fall in eastern Canada (Quebec and Ontario), eastern and south-eastern United States, and throughout the year in the Caribbean and South America.^(1,2,3)

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Reference 4 applies to this section.

Eastern Equine Encephalitis (EEE) is a member of the *Togaviridae alphavirus* family (EEE, Western Equine Encephalitis [WEE]). EEE can not survive outside of the host and is sensitive to drying, as well as moist and dry heat.

Clinical Presentation

EEE is the most severe of the arboviral encephalitides. It is an acute inflammatory disease of short duration involving the brain, spinal cord and meninges. An individual may develop symptoms four to 10 days after the bite of an infected mosquito. Mild cases often occur as a febrile headache or aseptic meningitis. More severe cases begin with the sudden onset of fever, myalgias, and headache of increasing severity. Many individuals will progress to more severe symptoms such as seizures and coma; those who recover may suffer permanent brain damage. The overall case fatality rate is approximately 35%.⁽⁵⁾

Diagnosis

Diagnosis of EEE is made through rapid serologic assays such as EIA. In early infection IgM antibody is more specific, while later in infection IgG antibody is more reactive. Viral isolation and identification have been useful in defining viral agents in serum, CSF and mosquito vectors.

Treatment

- Antibiotics are not effective.
- No effective antiviral treatment has been identified.

Reservoir

The true reservoir of EEE is unknown. It is thought that the virus may overwinter in birds or other animals. Emus are highly susceptible to EEE. Horses and humans are uncommon sources of mosquito infection.⁽⁵⁾

Transmission

EEE virus occurs in natural cycles involving birds and *Culiseta melanura* in some swampy areas almost every year during the warm months. How it survives in the cooler temperatures is unknown, and it reappears in the spring in birds and mosquitoes. EEE may also be transmitted by vectors such as *Coquilletidia perturbans* and *Aedes sollicitans*. These species feed on both birds and mammals and can transmit the virus to humans, horses and other hosts. Other mosquito species such as *Ae. vexans* and *Culex nigripalpus* can transmit EEE virus.

Incubation Period

The incubation period for EEE is usually five to 15 days.

Period of Communicability

EEE is not directly transmitted from person to person. The virus is not demonstrable in blood or CSF after onset of disease. In birds, the viremia lasts two to five days. Mosquitoes are infected for life (one to four months) and vertical transmission (female to egg) may exist. Viremia in horses is rarely present in high titres for long periods.

Host Susceptibility

Residents of and visitors to endemic areas, as well as people who engage in outdoor work and recreational activities, are at greater risk of infection with EEE. Persons over 50 and under 15 years of age seem to be at greatest risk for developing severe disease.⁽⁶⁾

Incidence

General

EEE is a summertime disease, occurring most frequently in children and the elderly.⁽⁷⁾ It occurs in eastern and north central United States (US) and Canada, in scattered areas of Central and South America, and in the Caribbean.⁽⁸⁾ In most regions of the US where EEE is enzootic, transmission to humans usually occurs between May and October.

Human cases occur relatively infrequently, largely because the primary transmission cycle takes place in swamp areas where populations tend to be limited; however, small outbreaks of human disease occur in the US. Epidemics of EEE are cyclic, with an interval between epidemics of about nine years.⁽⁶⁾ Human cases are usually preceded by cases in horses.

Canada

EEE has been isolated in Ontario, Alberta and Quebec.⁽⁶⁾

In 2001, an embargo was placed on shipments of Lucky Bamboo imported from Asia that arrived in standing water. The tiger mosquito, *Aedes albopictus*, had been identified as arriving with these shipments and was a potential source of EEE. This mosquito can transmit many viral agents which are not normally indigenous to Canada but are potentially harmful to humans and animals. In addition, this mosquito can transmit viral infections that already exist in Canada, such as EEE and WEE.⁽⁹⁾

EEE is not a nationally notifiable disease and the annual incidence is not known.

Alberta

There were no cases of EEE reported in Alberta between 1994 and 2020.

Public Health Management

Key Investigation

- Identify the potential source of the infection based on:
 - history of travel to endemic areas,
 - mode of transmission and
 - incubation period.
- Assess for history of mosquito bites.

Management of a Case

- Supportive care, including IV fluids and ventilation.
- Symptomatic treatment.

Management of Contacts

The infection is not passed from person to person.

Preventive Measures

- Educate the public about personal protective measures including:
 - reducing time outdoors in early evening hours,
 - wearing long pants and long-sleeved shirts, and
 - applying mosquito repellent containing DEET.
- Spray insecticides to kill larvae and adult mosquitoes.
- Immunize domestic animals (in endemic areas) and/or house them away from human living quarters.
 - Equine vaccine is available for EEE and WEE.⁽⁴⁾
- Support surveillance programs to allow for rapid identification of dangerous viruses in mosquito populations; this can shorten public health response time and reduce the spread of infected vectors.
- Laboratory safety – Arboviruses require biosafety level 3 practices.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links
	Incidence	• Updated Alberta Incidence

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